

Exposome / EHR rebuild: what needs your review

Branch `feature/decision-model-rebuild`, `jeonglab-bcm/exposome-ehr-review`

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Summary

The repository was rebuilt on the POTS-phenotyping design: one broad PubMed query, facets as tags rather than filters, and literature links found by decision models and confirmed by Claude with verbatim quotes. The code is tested (132 offline tests) and the counts below are taken from the committed harvest. What only you can decide are the four choices below. The model runs are built on Decisions 1 and 2, so changing those afterwards means re-running; review them first.

	Decision	What to look at	Section
1	Scope of the search	4 query blocks; what is deliberately left out; 4 known misses	1
2	Seed papers	8 papers the evidence graph is first built from	2
3	Vocabulary	67 terms in 6 groups; 9 flagged judgement calls	3
4	Per-paper questions	8 fields replacing the old checklist	4

Corpus: 15,913 articles, 6,951 tagged pediatric (prenatal, infant, child or adolescent). Section [5](#) lists what does not need your review.

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1 Decision 1: scope of the search

A paper enters the corpus if it matches any of the four blocks below (`config/query.yaml`). Nothing else is filtered at search time: age, publication type and open access are recorded per paper and filtered later.

Block	What it retrieves	Papers	Only here
exposome	The exposome concept itself (Wild 2005)	3,123	2,892
ewas	Agnostic, many-exposure association designs (Patel 2010) whose abstracts do not always say "exposome".	297	113
environment_in_health_records	The core of the old Tiers 2-4: an environmental or social exposure, studied in routinely collected individual-level data	11,624	11,558
vaccine_as_exposure	Old Tier 5: a vaccine as the exposure and a health event as the outcome, in surveillance, registry, claims or cohort data	1,116	1,109

Deliberately left out.

- **Occupational exposure.** PubMed’s “Environmental Exposure” heading includes it; 1,647 papers came in only that way, so the heading is used without its children (maternal, prenatal and inhalation exposure are kept).
- **Research-cohort studies with no routinely collected data**, unless they say “exposome” or are exposome-wide (e.g. a single chemical in one birth cohort). The old pipeline included some of these.
- **Vaccine uptake, coverage and hesitancy** studies (vaccination as the outcome, not the exposure); excluded by title only.

Known misses and the widenings declined. Of the old pipeline’s 30 EHR-positive papers, 11 are retrieved; 15 of the other 19 are out of scope (2 corrections, 10 programme reports, descriptive surveillance or attitude surveys, 2 research cohorts, 1 economic estimate). Four are real misses:

- PMC4997457 (air pollution and asthma exacerbations, Quebec administrative data): the abstract names only “hospitalization or emergency room visit”. Adding such phrases grows the corpus to about 19,800, mostly aggregate time-series studies.
- PMC8986305 (vaccine safety comparison): adding surveillance and pharmacovigilance terms to the vaccine block recovers it but adds about 1,600 papers.
- PMC4381500 (mortality after DTP/MMR eligibility) and PMC4113360 (prenatal bereavement, registry data).

Decision. Are these the right boundaries? In particular: include occupational exposure? include single-exposure research cohorts? accept the emergency-visit widening (+3,900 papers) or the vaccine-surveillance widening (+1,600)?

☐ Approve as is ☐ Change (note below) ☐ Discuss

Notes: _____

2 Decision 2: seed papers

The first model run reads these papers paragraph by paragraph; their links start the evidence graph, as POTS’s seven seeds did. Only CC BY text may be sent to the hosted Jev model, so most seeds rely on the local models agreeing.

Seed	PMID	Role	Licence	Paragraphs
wild2005	16103423	Defines the exposome concept (commentary)	abstract only	0
patel2010	20505766	Agnostic EWAS method, NHANES, adults	CC BY	49
maitre2018	30206078	HELIX: pediatric multi-exposure birth cohort	CC BY-NC	51
vrijheid2020	32579081	Pediatric exposome-wide study of obesity	EHP	58
brokamp2018	29126118	Geocoding environmental exposures onto EHR	CC BY-NC	32
hu2023	37089437	EHR exposome-wide study (OneFlorida)	CC BY-NC	28
correa2022	35970309	Neighbourhood factors and childhood asthma	author manuscript	33
macdonald2014	24914115	Vaccine as exposure, linked administrative data	abstract only	4

Wild 2005 has no abstract in PubMed and no PMC full text, so it contributes no paragraphs; it stays a seed because it defines the concept. The vocabulary’s `source_refs` were assigned from the seeds’ abstracts, not yet from reading them.

Decision. Keep these eight? Candidates to add: a pediatric EHR study of a chemical exposure (lead from EHR screening results), a Vaccine Safety Datalink study with an open licence, and an adult EHR exposome-wide study other than Hu 2023.

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Notes: _____

3 Decision 3: vocabulary

Each term is a tag, not a category: a paper carries any number. “Papers” is how many corpus papers carry the tag by either path (title/abstract query or NLM MeSH indexing). Rows marked ? are judgement calls.

3.1 Exposure

An environmental, chemical, social or medical factor a person encounters, studied as a possible cause of a health outcome.

Term	Definition (first sentence)	Papers
Particulate matter air pollution	Ambient fine or coarse particles (PM2.5, PM10, black carbon, ultrafine particles), usually assigned from a residential address	1,318
Traffic-related air pollution	Nitrogen dioxide, nitrogen oxides and other markers of road traffic, including distance to a major road and traffic density	738
Ozone	Ground-level ozone, a secondary photochemical air pollutant	430
Wildfire smoke	Smoke from wildfires, bushfires or landscape fires, as an episodic air pollution exposure	126
Indoor air and household fuel	Indoor sources: gas stoves, household solid fuel, mould and dampness, indoor allergens and radon	276
Tobacco smoke	Active smoking, maternal smoking in pregnancy, or second-hand (environmental) tobacco smoke, including e-cigarette aerosol ? <i>Active and second-hand smoking together.</i>	872

Term	Definition (first sentence)	Papers
Lead	Lead from paint, dust, water or soil, usually measured as blood lead concentration in children	249
Mercury, arsenic, cadmium and other metals	Toxic metals and metalloids other than lead (mercury, arsenic, cadmium, manganese), measured in blood, urine, hair or water	464
Phthalates, phenols and other endocrine disruptors	Non-persistent endocrine-disrupting chemicals: phthalates, bisphenol A and other phenols, parabens, triclosan	416
PFAS and persistent organic pollutants	Persistent chemicals: per- and polyfluoroalkyl substances, PCBs, organochlorine pesticides such as DDT/DDE, dioxins, flame retardants	568
Pesticides	Current-use pesticides: organophosphates, pyrethroids, herbicides such as glyphosate, and agricultural pesticide application near the home	931
Drinking water contamination	Contaminants in drinking water: nitrate, disinfection by-products, lead service lines, fluoride and water system violations	209
Ambient temperature and heat	Outdoor temperature, heat waves, cold spells and humidity, as a short-term or long-term exposure	422
Environmental noise	Road, rail and aircraft noise at the residence or school	137
Greenness and blue space	Vegetation and water near the home: NDVI greenness, parks, tree canopy, access to blue space	302
Built environment	The physical form of the neighbourhood: walkability, land use, building density, food outlets, urbanicity	1,259
Neighbourhood deprivation and opportunity	Area-level socioeconomic disadvantage measured by a composite index (Area Deprivation Index, Child Opportunity Index, Social Vulnerability Index, census-tract poverty) <i>? Area-level SDOH is separate from household SDOH (next row). Keep split?</i>	2,446
Household social and economic factors	Individual or household social determinants: income, insurance type, parental education, food insecurity, housing instability, homelessness	3,840
Structural racism and segregation	Residential segregation, historical redlining, discrimination and other structural measures of racism	120
Violence and crime exposure	Neighbourhood violent crime, gun violence, community violence and adverse childhood experiences	75
Vaccination	Receipt of a vaccine, its type, dose or timing, studied as an exposure that may precede an adverse event or another health... <i>? One facet for all vaccines. Split by vaccine type?</i>	1,324
Medication use in pregnancy or early life	A medicine taken in pregnancy or infancy studied as an exposure: antibiotics, acid suppressants, acetaminophen, antidepressants <i>? Only medicines in pregnancy or infancy. Adult drug exposures excluded.</i>	1,681

3.2 Health outcome

A disease, diagnosis, measurement or event studied as the consequence of an exposure.

Term	Definition (first sentence)	Papers
Asthma and wheeze	Asthma incidence, prevalence, exacerbation, emergency visit or hospitalisation, and recurrent or persistent wheeze	766

Term	Definition (first sentence)	Papers
Lung function	Spirometric or other lung function measurements: FEV1, FVC, FEV1/FVC.	169
Allergic disease	Atopic dermatitis, food allergy, allergic rhinitis and allergic sensitisation	219
Obesity and adiposity	BMI, BMI z-score, overweight, obesity, waist circumference or body fat, including rapid infant weight gain	998
Cardiometabolic risk	Blood pressure, hypertension, insulin resistance, type 2 diabetes, dyslipidaemia and metabolic syndrome <i>? Blood pressure, type 2 diabetes and lipids in one facet.</i>	931
Type 1 diabetes and autoimmune disease	Type 1 diabetes, coeliac disease, juvenile idiopathic arthritis, inflammatory bowel disease and other autoimmune conditions	371
Neurodevelopment and cognition	Cognitive function, IQ, developmental delay, language and motor development, school performance	809
ADHD and autism	Attention-deficit/hyperactivity disorder and autism spectrum disorder diagnoses or symptom scores <i>? ADHD and autism share one facet. Split?</i>	343
Mental and behavioural health	Depression, anxiety, behavioural problems, self-harm, substance use disorder and psychiatric emergency visits	1,161
Preterm birth and gestational age	Birth before 37 weeks, gestational age at birth, and spontaneous preterm labour	956
Birth weight and fetal growth	Birth weight, low birth weight, small or large for gestational age, and fetal growth measured by ultrasound	919
Congenital anomalies	Birth defects: congenital heart defects, neural tube defects, orofacial clefts and others	516
Pregnancy complications	Maternal outcomes of pregnancy: hypertensive disorders including preeclampsia, gestational diabetes, stillbirth and severe maternal morbidity	419
Childhood cancer	Cancer in children and adolescents, chiefly leukaemia, brain tumours and lymphoma <i>? Only childhood cancer. Adult cancers are tagged nowhere.</i>	1,083
Infection and infection-related care	Infections and their consequences: respiratory infections, COVID-19, bronchiolitis, otitis media, infection-related hospitalisation	1,973
Vaccine adverse event	An event studied as a possible adverse reaction to a vaccine: febrile seizure, fever, anaphylaxis, Guillain-Barre syndrome, intussusception, myocarditis	466
Mortality	Death from any cause or a specific cause, including infant mortality and sudden infant death <i>? All-cause and cause-specific mortality together.</i>	2,748
Health care use	Emergency department visits, hospitalisation, primary care visits or costs, studied as an outcome <i>? Health care use as an outcome, not as a data source.</i>	1,778
Puberty timing and linear growth	Age at puberty or menarche, precocious puberty, and height or linear growth	74

3.3 Data source

Where the individual-level data came from.

Term	Definition (first sentence)	Papers
Electronic health records	Electronic health or medical records from clinical care: diagnoses, encounters, vital signs, laboratory results, medications	2,348

Term	Definition (first sentence)	Papers
Insurance claims	Billing or insurance claims data: Medicaid, Medicare, commercial claims, national health insurance databases	964
Health registry	A population or disease registry: birth, death, cancer, prescription, patient or national health registers	6,040
Immunization registry or vaccine surveillance system	An immunization information system or vaccine safety surveillance system: Vaccine Safety Datalink, VAERS, provincial vaccine registries	440
Record linkage	Individual-level linkage of two or more data sources, such as hospital records to a birth registry or to environmental data by address	443
Research cohort	Data collected for research by a prospective cohort, most often a birth or mother-child cohort with its own visits, samples and questionnaires	2,914
National health survey	A repeated cross-sectional national survey with examination or biomarker data, such as NHANES.	305
Biobank	A biobank linking stored samples or genotypes to health records, such as UK Biobank or All of Us	279

3.4 Exposure assessment method

How an exposure was assigned to a person: from an address, a model, a monitor, a biological sample, a questionnaire or a record.

Term	Definition (first sentence)	Papers
Residential address geocoding	Assigning area-level exposures from a geocoded home address, census tract or postcode	1,332
Spatial or satellite exposure model	An exposure surface estimated by land-use regression, dispersion, chemical transport, satellite remote sensing or machine-learning models	311
Fixed monitoring station	Exposure taken from the nearest regulatory air quality or weather monitor	583
Biomonitoring	Measuring a chemical or its metabolite in blood, urine, hair, breast milk or another biological sample	1,324
Questionnaire or interview	Exposure reported by the participant or a parent in a questionnaire or interview	2,062
Personal sensor or wearable	A personal monitor, wearable sensor, smartphone or GPS tracking used to measure an individual's exposure directly	243
Exposure taken from health records	An exposure read from the health record itself: a vaccination record, a prescription, a recorded smoking status or a screening test result	298

3.5 Study design or analysis

The design or analytic approach.

Term	Definition (first sentence)	Papers
Exposome-wide association design	Testing many exposures agnostically against one outcome with correction for multiple testing (EWAS, ExWAS), including polyexposure risk scores	948
Exposure mixture methods	Statistical methods for joint effects of correlated exposures: weighted quantile sum, Bayesian kernel machine regression, quantile g-computation	229

Term	Definition (first sentence)	Papers
Self-controlled design	A design in which each person is their own control: self-controlled case series, case-crossover, risk-interval designs	415
Time-series design	Daily or weekly counts of events regressed on short-term exposure variation, often with distributed lags	331
Machine learning	Random forests, gradient boosting, neural networks, LASSO or other machine learning used for prediction or variable selection	554
Gene-environment interaction	Joint analysis of genotypes and exposures, including GxE interaction tests and polygenic-by-exposure models	300

3.6 Age stratum

The life stage the study enrolled or the exposure window it studied.

Term	Definition (first sentence)	Papers
Pregnancy or prenatal window	Exposure in pregnancy, or pregnant people enrolled	3,697
Infants	Newborns and infants under two years old	3,236
Children	Children from two to twelve years old	4,356
Adolescents	Young people from thirteen to eighteen years old	2,629
Adults	Adults, including older adults	7,290

Not in the vocabulary. Diet and nutrition, physical activity, sleep, the microbiome, ionising radiation, occupational exposures, adult cancers, and kidney or liver outcomes. Adding any of them is a new facet in `config/facets.yaml`; papers are re-tagged without re-searching.

Decision. Approve the term lists? Mark terms to split, merge, rename or add. The flagged rows above are the ones I was least sure of.

☐ Approve as is ☐ Change (note below) ☐ Discuss

Notes: _____

4 Decision 4: per-paper questions

These replace the old free-text checklist (`config/study_coding.yaml`). Each is a yes/no or a fixed set of options, answered from the passages listed, with the probability and the paragraph that carried the answer kept. Summary, key findings and limitations are no longer produced: they are not decisions.

Field	Type	Reads	Options
<code>primary_research</code>	yes no	abstract	yes / no
<code>routinely_collected_data</code>	yes no	abstract, methods	yes / no
<code>data_sources</code>	any of	abstract, methods	the data source terms of Section 3
<code>age_strata</code>	any of	abstract, methods	the population terms of Section 3
<code>study_design</code>	one of	abstract, methods	cohort; case_control; cross_sectional; self_controlled; time_series; trial; exposome_wide
<code>exposures_studied</code>	any of	abstract	the exposure terms of Section 3
<code>outcomes_studied</code>	any of	abstract	the outcome terms of Section 3

Field	Type	Reads	Options
<code>data_availability</code>	one of	availability, methods	<code>public_repository</code> ; <code>on_request</code> ; <code>restricted</code> ; <code>supplementary_only</code>

`routinely_collected_data` keeps the old `ehr_used` definition: electronic, individual-level and collected for care, billing or administration before the study. Chart review by study staff, research questionnaires and aggregate counts do not count. The pediatric EHR corpus is then: primary research = yes, routinely collected data = yes, and age strata include infant, child or adolescent.

Decision. Are these the fields your review needs? Anything to add (e.g. sample size band, country, outcome definition by codes vs. clinical measure, exposure window)? Is the `ehr_used` definition still right?

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Notes: _____

5 What does not need your review

- The harvester, parser, store and concept resolvers: ported from POTS-phenotyping with their tests.
- The decision-model and Claude-judge code: tested offline with a fake model; also run end to end over the 255 seed paragraphs.
- Derived tables (`data/derived/`) and the harvest report: generated; 24 of 26 checks pass. The two failures are 19 terms with no OMOP concept match, mostly exposure concepts that the free SNOMED lookup does not cover; an ATHENA vocabulary download fixes most of them.
- The move of the old pipeline to `legacy/`.

After your review

1. Apply your changes to the config files; re-harvest only if Decision 1 changed.
2. Run the seed reader run on the lab server and the Claude judge (`docs/DECISION_MODELS.md`).
3. Hand-code 30–50 papers for the Decision 4 fields to measure agreement before the coding is used.
4. Before merging: the GitHub Pages site (old site moved to `legacy/docs/`), and the POTS branch this is built on, which is not yet merged.

A Link predicates

Predicate	From → to	Question option
<code>risk_factor_for</code>	exposure → outcome	an exposure associated with more of, or a higher risk of, an outcome
<code>protective_for</code>	exposure → outcome	an exposure associated with less of, or a lower risk of, an outcome
<code>no_association_with</code>	exposure → outcome	an exposure found not to be associated with an outcome
<code>assessed_by</code> <code>ascertained_from</code>	exposure → assessment outcome, exposure → data_source	how an exposure was measured or assigned the data source an outcome or exposure was taken from
<code>linked_with</code>	data_source, assessment → data_source, assessment	two data sources joined person by person, or by address